Cambridge Assessment International Education

Cambridge International AS Level

MARINE SCIENCE

9693/02

Paper 2 AS Level Data-handling and Investigative Skills

MARK SCHEME

Maximum Mark: 75

Specimen Paper for examination from 2022

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

This document consists of 5 printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:
• the specific content of the mark scheme or generic level descriptors
• the specific skills defined in the mark scheme
• the standard of response required by candidate scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always whole marks (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded positively for valid answers that clearly demonstrate knowledge and skills.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently across all candidates.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme.

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on requirements as defined in the mark scheme.

Science-Specific Marking Principles & Abbreviations

Abbreviations: ; separates marking points | ***/*** alternative answers | **R** reject | **A** accept | **AW** alternative wording | **underline** essential word | **max** maximum mark | **ecf** error carried forward | **I** ignore.

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Question	Answer	Marks
1(a)	correctly plotted points on line graph ; appropriate scale chosen using >50% grid on both axes ; line drawn with ruler point-to-point / line of best fit ; *A ±1 small square tolerance*	3
1(b)	coral calcification decreases as temperature exceeds 30 °C ; thermal stress leads to expulsion of zooxanthellae / coral bleaching ; loss of photosynthetic products (glucose / amino acids) ; reduced energy available for calcium carbonate skeleton deposition ; *A ocean acidification effect*	4
1(c)	constant light intensity / photoperiod ; constant water pH / dissolved oxygen ; constant species / colony size of coral ; use water baths with thermostatically controlled heaters / chillers ; *max 4 for valid controlled variables*	4
1(d)	repeat at each temperature at least 3 times ; calculate mean calcification rate to reduce anomaly impact ; test smaller temperature increments between 28 °C and 32 °C to pinpoint threshold ; *A standard deviation / error bars*	3
2(a)	haemocytometer counting chamber used ; count cells in standard grid squares using inclusion rule (top/left) ; calculate concentration using chamber depth (0.1 mm) and grid area ; *A optical density / spectrophotometer*	3
2(b)	light penetration decreases exponentially with depth due to absorption/scattering ; compensation depth: rate of photosynthesis equals rate of respiration ; net primary production is zero at compensation depth ; below compensation depth, respiration exceeds photosynthesis leading to net biomass loss ; *A euphotic zone limit*	4
2(c)	nutrient limitation (nitrate / phosphate) in surface waters ; presence of thermocline prevents upwelling / mixing of nutrient-rich deep water ; zooplankton grazing pressure reduces standing crop ;	3

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Question	Answer	Marks
2(d)	calculate percentage increase = [(final - initial) / initial] × 100 ; substitution: correct numerical values from Table 2.1 ; correct final value with % sign ; *ecf for arithmetic error*	3
3(a)	marine teleost body fluids are hypotonic to seawater ; continuous osmotic water loss across permeable gill surfaces ; inward diffusion of sodium and chloride ions down concentration gradient ;	3
3(b)	drinks seawater continuously to replace lost water ; active transport / excretion of excess Na+ and Cl- by chloride cells in gills ; minimal volume of highly concentrated / isotonic urine produced by kidneys ; magnesium and sulfate ions actively secreted in urine / feces ;	4
3(c)	freshwater teleost fluids are hypertonic to surrounding freshwater ; water enters by osmosis across gills; does not drink water ; produces large volume of dilute urine; actively absorbs ions across gills ;	3
4(a)	primary producer: phytoplankton / seagrass ; primary consumer: zooplankton / herbivorous fish ; secondary consumer: carnivorous fish / squid ; tertiary / top predator: apex predator (shark / tuna) ; *A correctly oriented arrows showing energy flow*	4
4(b)	energy transfer efficiency = (energy in trophic level n+1 / energy in trophic level n) × 100 ; typically 10% (range 5-20%) transferred between trophic levels ; energy lost as heat from cellular respiration ; energy lost in uneaten parts, feces / excretion, and natural mortality ;	4
4(c)	pyramid of biomass for marine pelagic ecosystems can be inverted (temporary) ; phytoplankton have high turnover rate / rapid reproduction rate ; small standing biomass supports larger biomass of long-lived zooplankton ; pyramid of energy is always upright / obeys second law of thermodynamics ;	4
4(d)	calculate energy available to top predator = initial GPP × efficiency^steps ; correct calculation working shown ; final value with units (kJ m^-2 yr^-1) ; *ecf accepted*	3

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Question	Answer	Marks
5(a)	interrupted belt transect laid perpendicular to shoreline (low water to high water) ; place quadrats at regular intervals (e.g. every 5 m) along transect ; record percentage cover of sessile species / count individuals of mobile species ;	3
5(b)	physical desiccation / drying out at upper shore during low tide ; temperature fluctuations and salinity stress from rain / evaporation in rock pools ; wave action / mechanical hydrodynamic force at lower/middle shore ;	3
5(c)	interspecific competition for space / attachment substrate ; predation pressure higher at lower shore from marine predators (crabs, sea stars) ; upper limit determined by abiotic tolerance, lower limit determined by biotic interactions ;	3
6(a)	bioaccumulation: buildup of non-biodegradable substance within an individual organism over time ; biomagnification: increase in concentration of pollutant at progressively higher trophic levels ; pollutant is lipophilic / fat-soluble and not easily metabolized or excreted ;	3
6(b)	microplastics (<5 mm) ingested by filter feeders mistaking them for plankton ; cause physical blockage of digestive tract, leading to pseudo-satiation and starvation ; toxic chemicals (PCBs, heavy metals) adsorb onto plastic surface ; plasticizers (phthalates, BPA) leach into tissues acting as endocrine disruptors ;	4
6(c)	describe trends from Fig. 6.1 (positive correlation between trophic level and concentration) ; cite specific data values from graph comparing primary consumer to apex predator ; calculate magnification factor between lowest and highest trophic level ; identify any anomalies or deviations from trend ;	4
6(d)	enforce international MARPOL regulations prohibiting disposal of plastics at sea ; ban primary microbeads in cosmetics and industrial abrasives ; implement catchment management and waste-water filtration traps at river mouths ;	3

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